MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-flavordb_full_r033/naveja_recap

generated 2026-08-20 07:06

A · Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333, 'include_ring': True}
core_ratio	0.3333
max_cores	10
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	2
rdkit_version	2026.03.2
created_at	2026-08-20T06:29:01

B · Composition

quantity	value
molecules	413,459
from coconut	391,965 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	3,455,836
per molecule	mean 8.36, median 10, max 10
R-group slots	3,851,767
per decomposition	mean 1.11
k >= 2	9.47% of decompositions

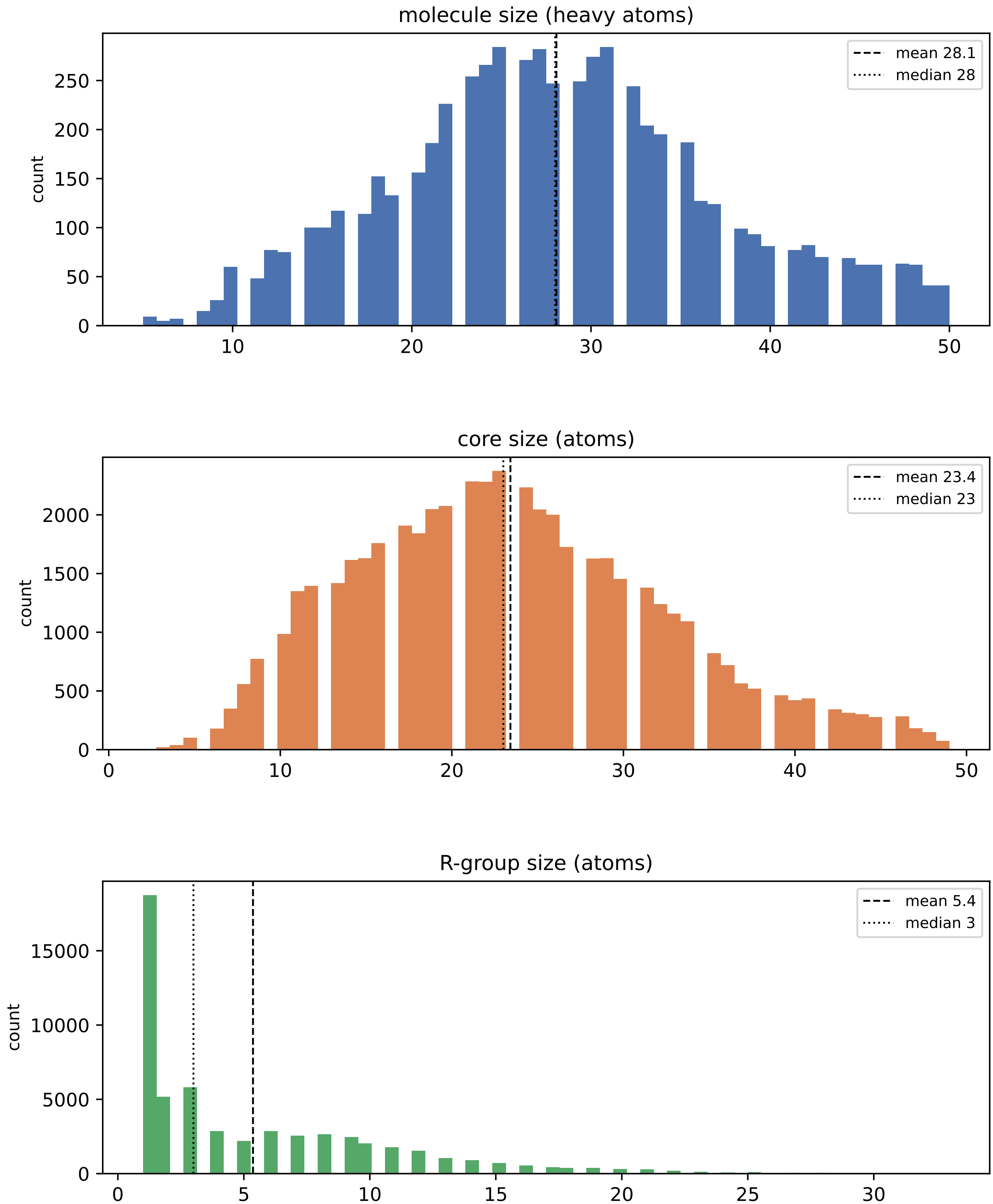
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	413,459 instances/epoch
sampling_unit = decomposition	3,455,836 instances/epoch (8.4x)
full (mol, core, islinked) space	4,493,636 (10.9 per molecule)

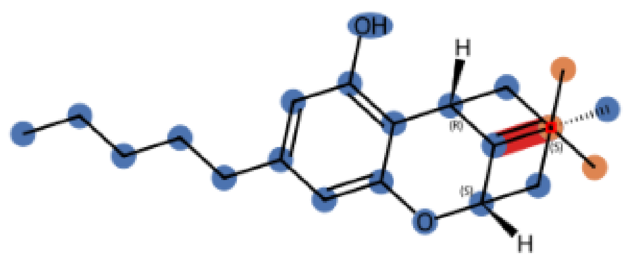
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

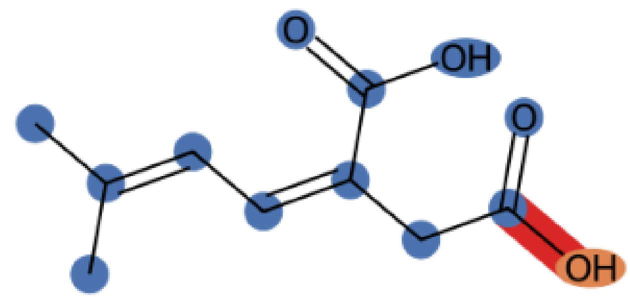


E · Example decompositions (ratio 0.3333)

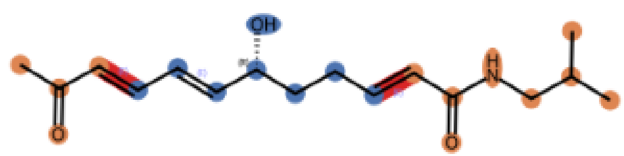
CNP0000074 k=1 core 20 atoms, R-groups [3]



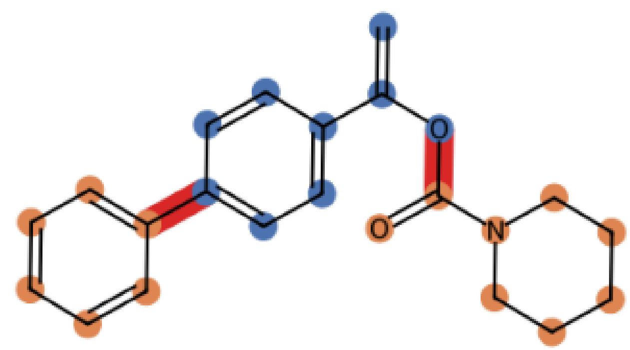
CNP0000161 k=1 core 12 atoms, R-groups [1]



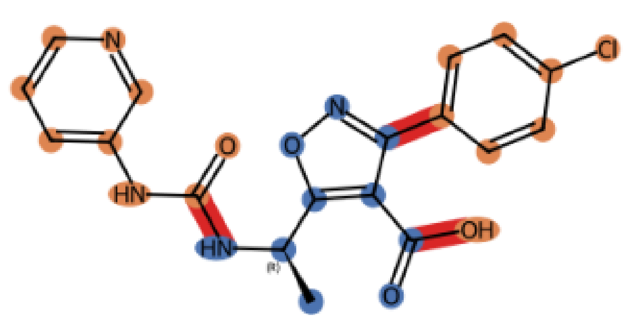
CNP0000307 k=2 core 8 atoms, R-groups [4, 8]



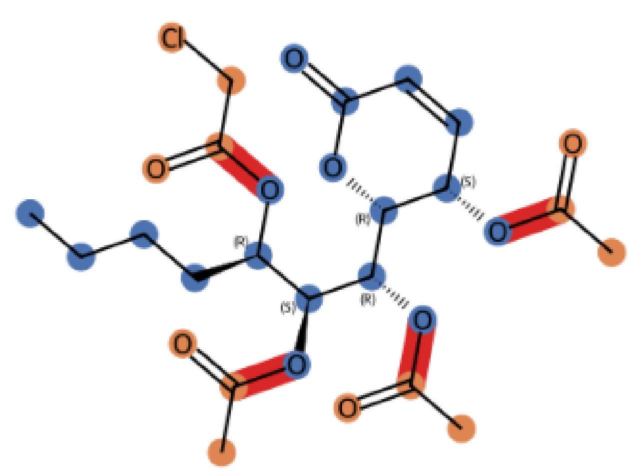
CNP0001292 k=2 core 9 atoms, R-groups [8, 6]



CNP0014826 k=3 core 10 atoms, R-groups [9, 7, 1]

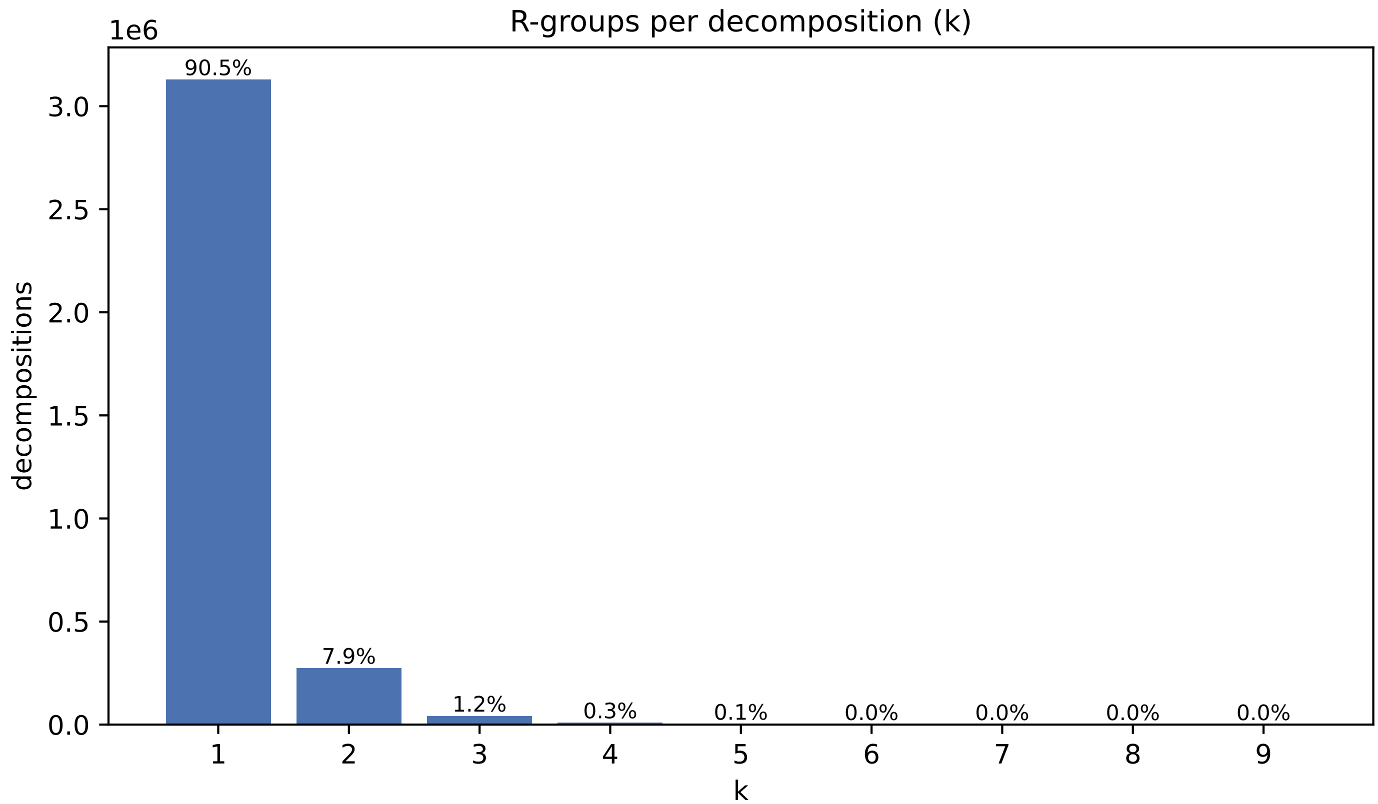


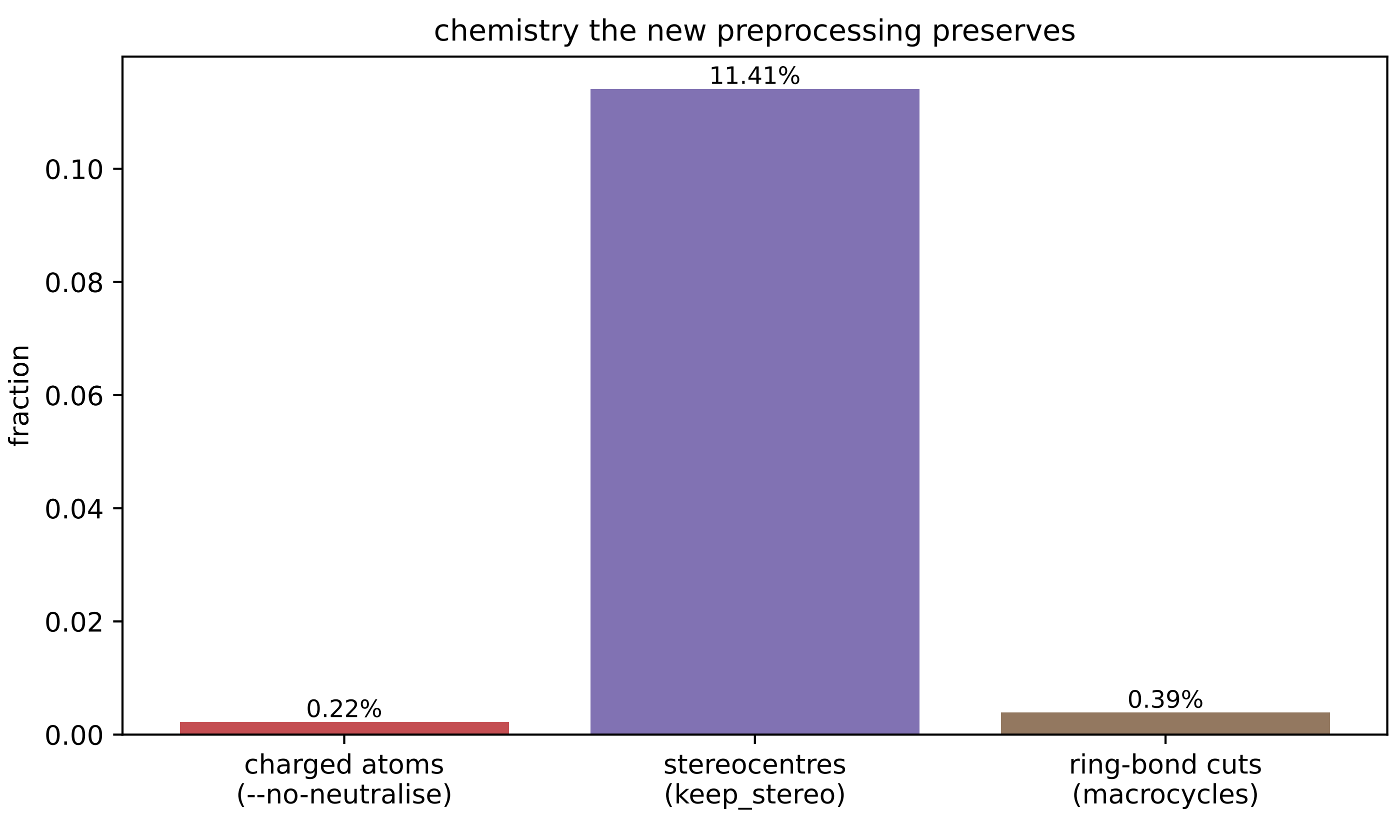
CNP0078333 k=4 core 18 atoms, R-groups [4, 3, 3, 3]



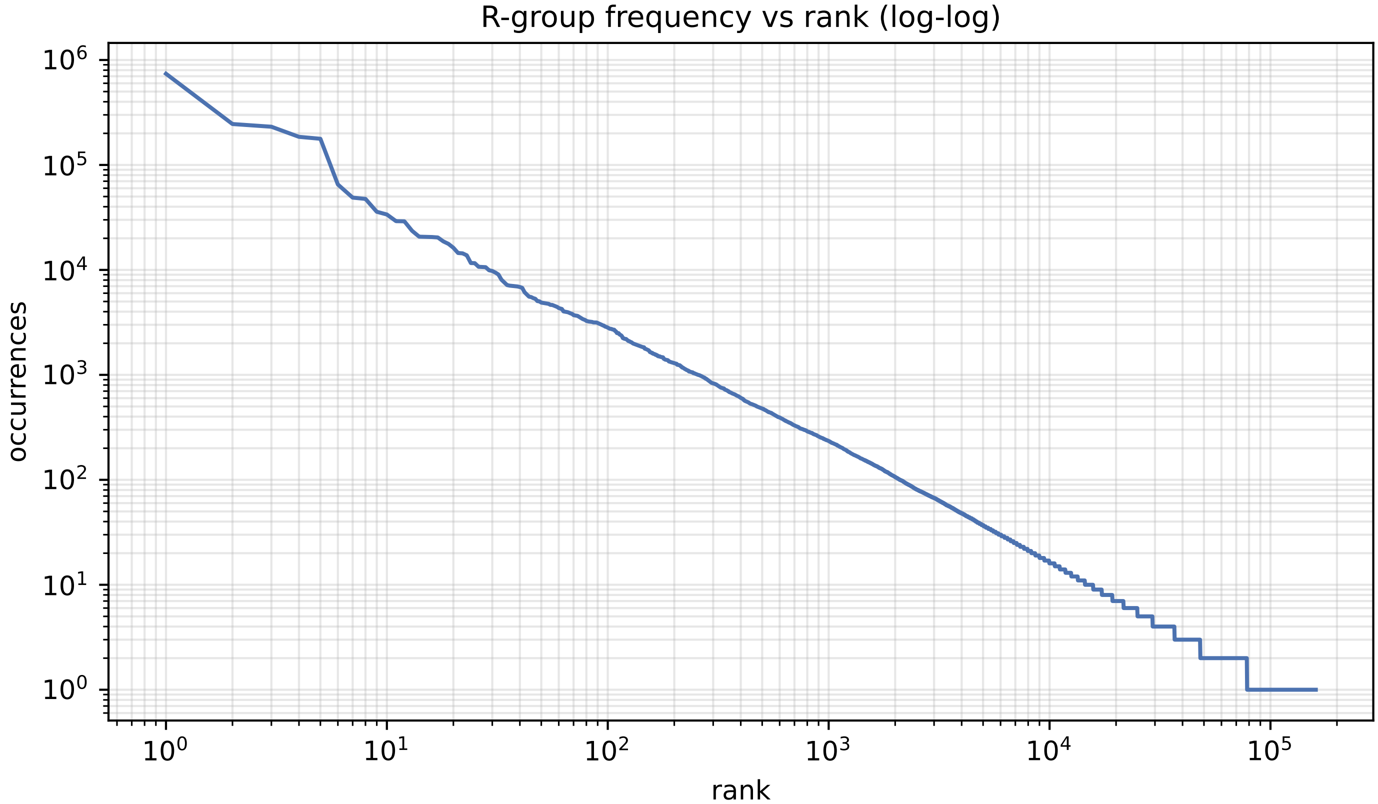
blue = core orange = R-groups red = cut bonds

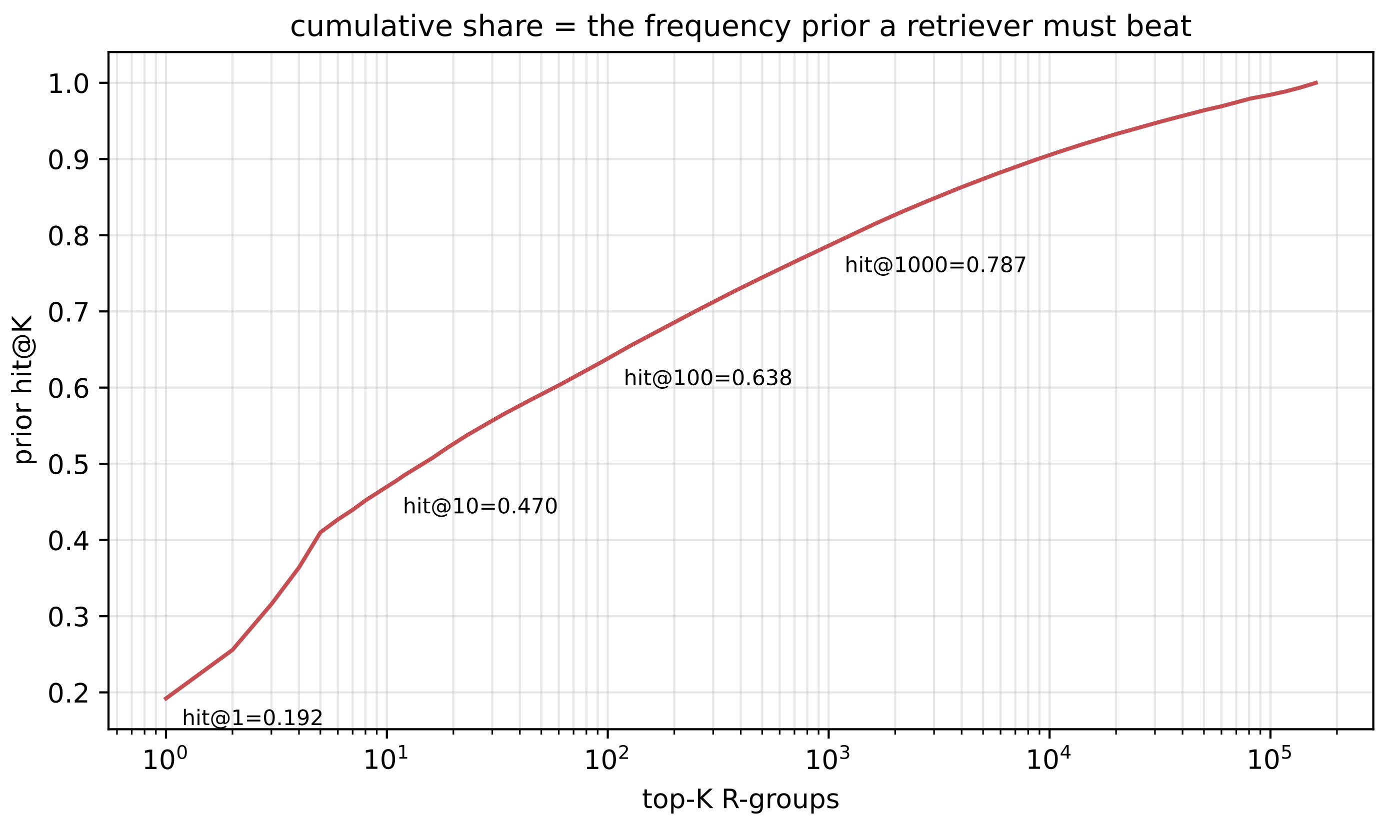
F · Decomposition shape & preserved chemistry





G · R-group library skew





H · R-group library

quantity	value
distinct R-groups	160,431
occurrences	3,851,767
effective size (exp H)	444
top-1 share	19.20%
top-10 cover	47.0%
top-100 cover	63.8%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
739,638	19.20%	*C
245,402	6.37%	*O
231,272	6.00%	*O
185,329	4.81%	*C(C)=O
177,543	4.61%	*OC
65,145	1.69%	*CC
48,829	1.27%	*CO
47,477	1.23%	*c1ccccc1
35,794	0.93%	*C(C)C
33,719	0.88%	*OC(C)=O
29,139	0.76%	*CCC
29,039	0.75%	*C(C)C
23,588	0.61%	*C(=O)O
20,729	0.54%	*CCCC
20,668	0.54%	*c1ccc(OC)cc1
20,569	0.53%	*Cl
20,380	0.53%	*OC
18,701	0.49%	*C(=O)OC
17,695	0.46%	*C(=O)c1ccccc1
16,249	0.42%	*OCC

J · Instance-level common-R-group filter

quantity	value
percentile	99.99
count threshold	20,308
hashes flagged common	17 of 160,431
their share of occurrences	51.3%
single-R-group instances rejected	29,068/56,364 (51.6%)
mean R-group size, rejected	1.75 atoms
mean R-group size, kept	9.21 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.